Introduction to R in 1 Hour

(Created 07/15/2020, Updated 07/11/2023)

This quick introduction to R is meant for students who have not used R before but are reasonably comfortable with computers. There are different ways to run R: this handout will introduce the RStudio Cloud in Section 1 and local RStudio on your desktop/laptop in Section 2. R will provide the backbone and RStudio will provide a better interface than the command terminal in R.

1. Posit Cloud (used to be called RStudio Cloud)

You can sign up at https://posit.cloud/ and start using Posit (used to be RStudio) and R right away. There is no need to install software and packages on a local machine. But the free accounts are limited in features (particularly the number of hours per month).

2. Download/install R and RStudio

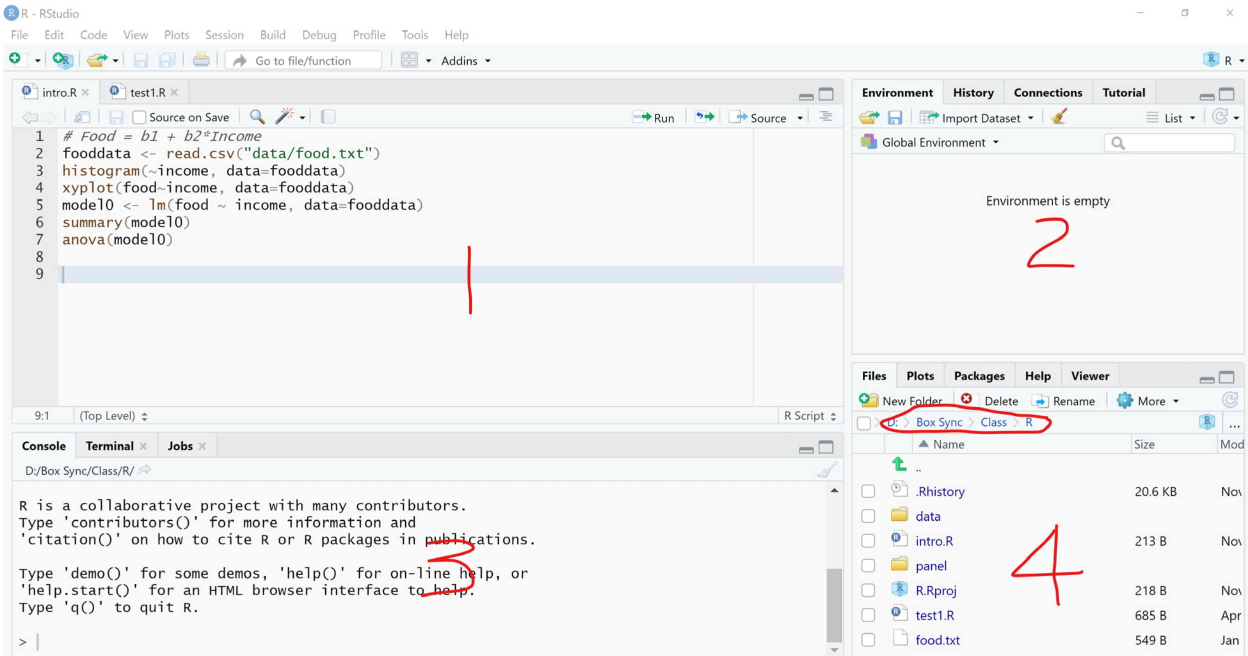
You only need to do this once on a computer. Steps to set up the software:

1. Go to https://www.r-project.org/ and click on "download R." This should bring you to the mirror sites and pick one that is geographically close to you. Depending on your OS, choose "Download R for Windows/Linux/Mac." If Windows, click on "install R for the first time." Download the installation file on top.
2. Run the downloaded setup file and install R.
3. Go to https://posit.co/ and click on "Download" and choose RStudio Desktop (Free), then download the installation file.
4. Run the downloaded setup file and install RStudio.
5. The above link descriptions may change over time. But it should be easy to locate the downloads.

Once you have installed both, you can click on the RStudio icon to start your first R session. RStudio will provide a seamless experience. You do not need to interact with R directly.

3. RStudio Interface

Look around in RStudio and make yourself comfortable with the R Studio interface.



There are roughly four windows as you can see. Windows 1 is the script window (which is similar to the Do-Editor in Stata): you can see your scripts/programs here. Windows 2 and 3 are pretty obvious. Window 4 is also obvious. Pay attention to the path I highlighted “D:\Box Sync\Class\R” – your window may show a different path/folder; that is okay, this is your current directory/folder; you want to unzip the script/data files into this folder. Or, you can browse to the folder where you saved the files and set it as your working directory.

4. First Interactive Session

In Window 3, type in “10*pi” without the quotation marks and you will see 31.41593. Now you can use R as a calculator. Type in

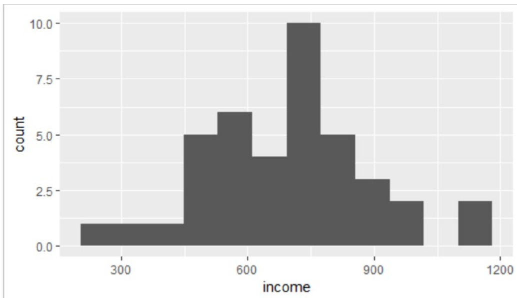
```
x <- 2
y <- x/5
x
y
```

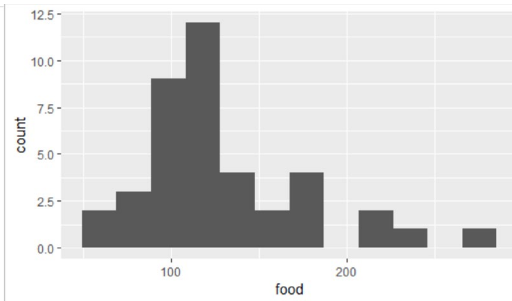
Now you will see that x is 2 and y is 0.4.

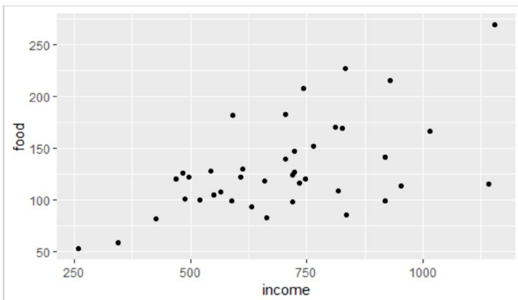
5. First Regression in a Script

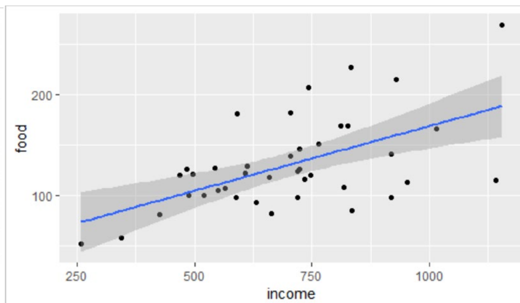
In R Studio, click on File, then Open file … and then choose the intro.R you just downloaded with this handout. This would open the script file as you see in Window 1. This script will do something similar to what we did in our first Stata lab session.

Click on Source on the upper right corner of Window 1 or press Ctrl-Shift-Enter (press Ctrl first, hold it, then press shift, hold both, then press Enter key) to run the script. This would produce four plots below.









Command “model <- lm(food ~ income, data=fooddata)” tells R that you want to estimate a linear model of food on income (plus an intercept by default), using the data fooddata.

In Window 3, you can see the regression output:

```
> summary(model)

Call:
lm(formula = food ~ income, data = fooddata)

Residuals:
    Min      1Q  Median      3Q     Max 
-71.753 -19.666  -5.969  17.752  80.140 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 40.76756   22.13865   1.841 0.073369 .  
income       0.12829    0.03054   4.201 0.000155 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 37.81 on 38 degrees of freedom
Multiple R-squared:  0.3171,	Adjusted R-squared:  0.2991 
F-statistic: 17.65 on 1 and 38 DF,  p-value: 0.000155

> anova(model)
Analysis of Variance Table

Response: food
```

```
          Df Sum Sq Mean Sq F value   Pr(>F)    
income     1  25221 25221.2  17.646 0.000155 ***
Residuals 38  54311  1429.2                     
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

You can compare the results here with the Stata output in our first lab session. Coefficients, standard errors, t-values, R-squared, overall sig F-stat, … are all identical.

Here you go – you just ran your first regression in R! To continue to learn R, I recommend the following two web pages:

https://bookdown.org/ccolonescu/RPoE4/

https://www.econometrics-with-r.org/

Obviously, there are plenty of books and videos on R available online. Have fun!